

Multi-species Catalogues of Regulatory Elements: Ensembl

Session 3

T6: Making the most of cis-regulatory annotations for genome-scale analysis

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03/09/2026

Session map

The Ensembl Project	• General description
Ensembl Regulation	• Regulatory pipelines • Available datasets and Regulatory annotation • Defining and accessing reference epigenome data
The Ensembl Regulation subsite	• Epigenomes data and metadata • The regulatory features tab • Exploring genome annotation and regulatory activity
Comparison of CREs frameworks	• Most-popular genome-wide resources • Strengths and weaknesses of each approach
Hands-on	

The Ensembl Project

The mission of Ensembl is to provide **comprehensive and accurate annotation** of genomes, **integrating** this information with other biological data and making it **publicly accessible** via the web.



ENSEMBL



Have you used Ensembl before?



Using genome browsers to visualise annotation

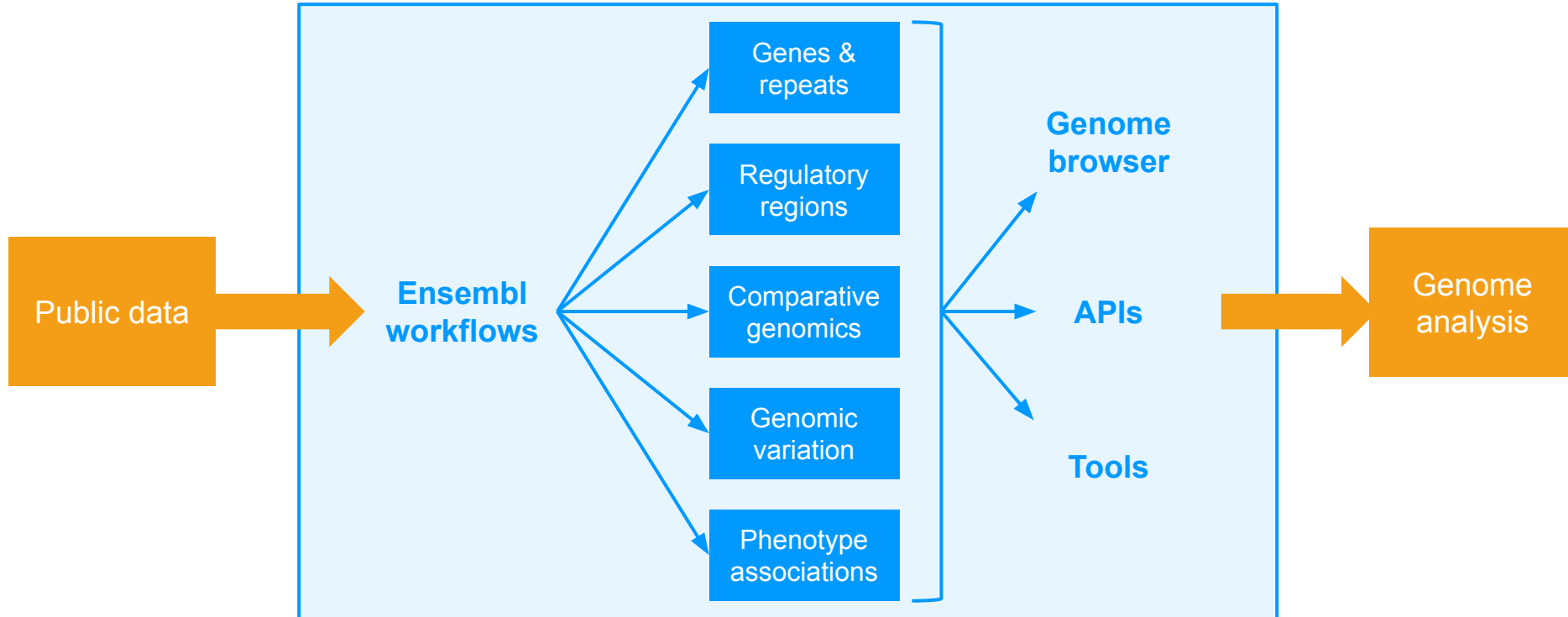
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Regulatory element

Gene

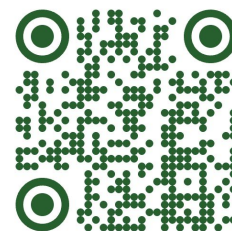
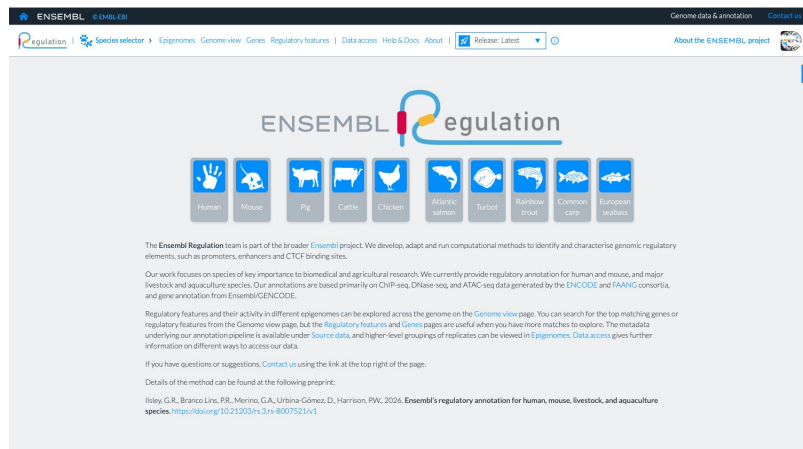
Variant

Ensembl enables genome analyses

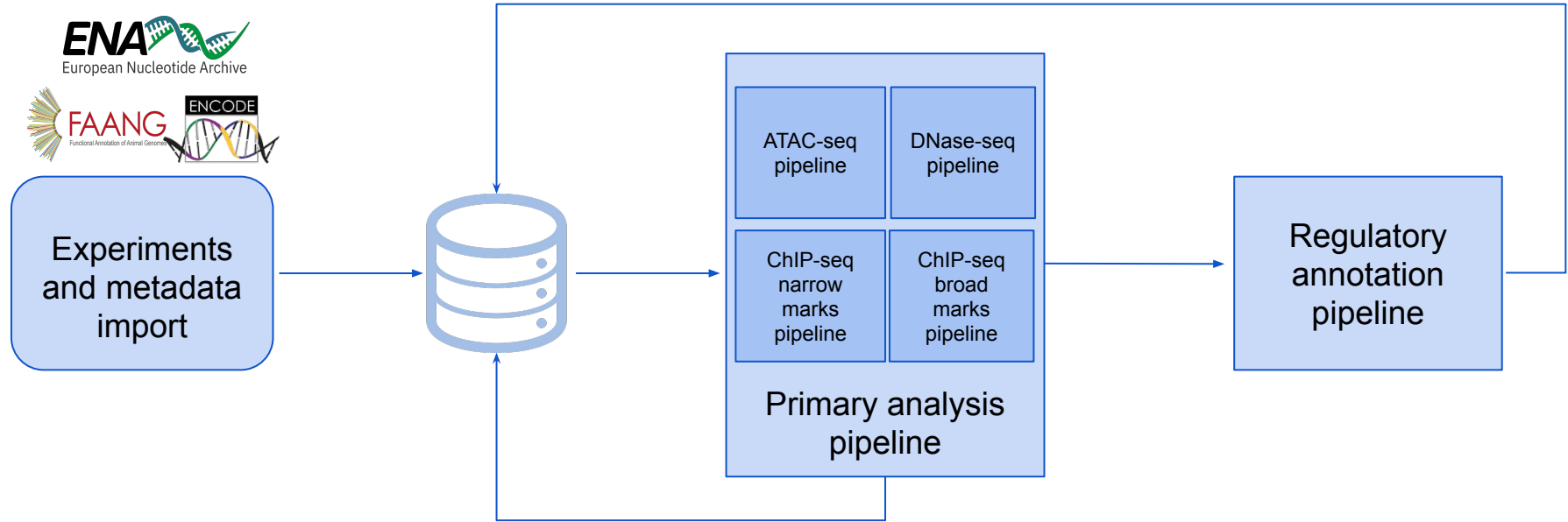


Ensembl Regulation

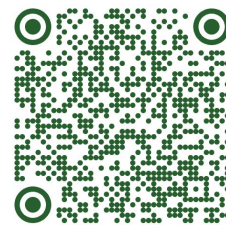
- ✓ Aim: Develop, adapt and run computational methods to identify and characterise genomic regulatory elements, such as promoters, enhancers and CTCF binding sites.
- ✓ Focus: Species of key importance to biomedical and agricultural research.
- ✓ Annotations are based primarily on ChIP-seq, DNase-seq, and ATAC-seq data generated by the ENCODE and FAANG consortia, and gene annotation from Ensembl/GENCODE.



The Ensembl Regulatory Build



Primary analysis pipeline



- ✓ Processes read files to produce peak (BED) and signal (bigWig) files.



- Preprocessing: Adapters trimming with NGmerge/fastp.
- Alignment: Bowtie2.
- Peak calling:
 - Genome mask:
 - ENCODE masks for human and mouse.
 - Custom masking strategy for other species.
 - Genrich with different configurations for each experiment type.
- Signal: Estimated from Genrich pileups

Standardised data analysis pipelines

- **ATAC-seq & DNase-seq** peaks for all the epigenomes of the target species
- H3K27ac, H3K4me1, and H3K4me3 ChIP-seq peaks for all the epigenomes of the target species (**optional**)
- Annotated TSSs and CDSs from transcripts of interest



- Set of regulatory features: **Promoters, Enhancers, OCR**
- Regulatory features (RFs) **activity** across different epigenomes with open **chromatin data**

The Ensembl Regulation Subsite

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Regulation



Species selector

Epigenomes

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Search...



Human (*Homo sapiens*)



Mouse (*Mus musculus*)



Pig (*Sus scrofa*)



Cattle (*Bos taurus*)



Chicken (*Gallus gallus*)



Atlantic salmon (*Salmo salar*)



Turbot (*Scophthalmus maximus*)



Rainbow trout (*Oncorhynchus mykiss*)



Common carp (*Cyprinus carpio carpio*)



European seabass (*Dicentrarchus labrax*)

ENSEMBL Regulation



Pig



Cattle



Chicken



Atlantic salmon



Turbot



Rainbow trout



Common carp



European seabass

Species selector

in computational methods to identify and characterise genomic regulatory

Our work focuses on species of key importance to biomedical and agricultural research. We currently provide regulatory annotation for human and mouse, and major livestock and aquaculture species. Our annotations are based primarily on ChIP-seq, DNase-seq, and ATAC-seq data generated by the [ENCODE](#) and [FAANG](#) consortia, and gene annotation from Ensembl/GENCODE.

The Ensembl Regulation Subsite: Epigenomes tab

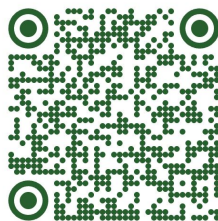
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Regulation Human > Epigenomes Genome view Genes Regulatory features | Data access Help & Docs About | Release: Latest About the ENSEMBL project

GRCh38

Showing 144 epigenomes Download

Epigenome	Biosample term	Tissue group	Sex	Life stage	Age	Biosample material	Assays
A549 (m, 58 y)	A549	lung	male	adult	58 y	cell line	
A673 (f, 15 y)	A673	muscle	female	child	15 y	cell line	
ACC112 (m, 70 y)	ACC112	epithelium	male	adult	70 y	cell line	
adrenal gland (f, 53 y)	adrenal gland	endocrine gland	female	adult	53 y	primary tissue	
adrenal gland (m, 37 y)	adrenal gland	endocrine gland	male	adult	37 y	primary tissue	
adrenal gland (m, 54 y)	adrenal gland	endocrine gland	male	adult	54 y	primary tissue	
adrenal gland (96-97 dpf)	adrenal gland	endocrine gland	male, unknown	embryonic	96-97 dpf	primary tissue	
AGO4450 (m, 12 w)	AGO4450	connective tissue	male	embryonic	12 w	cell line	



GRCh38 epigenomes / A549 (m, 58 y) Use epigenome

ATAC-seq 3 Biosamples Download

Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR032RGS	ENCB0003HWJ	cell line	A549	adult	male	58 years
ENCSR032RGS	ENCB0049MEE	cell line	A549	adult	male	58 years
ENCSR032RGS	ENCB0024PIA	cell line	A549	adult	male	58 years

DNase-seq 2 Biosamples Download

Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR000ELW	ENCB163AAA	cell line	A549	adult	male	58 years
ENCSR000ELW	ENCB888BLUL	cell line	A549	adult	male	58 years

ChIP-seq (CTCF) 1 Biosample Download

Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR000DNA	ENCB162AAA	cell line	A549	adult	male	58 years

ChIP-seq (H3K4me3) 2 Biosamples Download

Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR000DPD	ENCB888BLUL	cell line	A549	adult	male	58 years
ENCSR000DPD	ENCB163AAA	cell line	A549	adult	male	58 years



Epigenomes tab

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Regulation Human Epigenomes Genome view Regulatory features Source data Help & Docs About Release: Latest About the ENSEMBL project

GRCh38

Showing 4 epigenomes.

Epigenome	Biosample term	Tissue group	Sex	Life stage	Age	Biosample material	Assays
astrocyte	astrocyte			unknown		primary cell	
brain (120-122 dpf)	brain			embryonic	120-122 dpf	primary tissue	
neural progenitor cell (f, 5 dpf)	neural progenitor cell			embryonic	5 dpf	primary cell culture	
SK-N-SH (f, 4 y)	SK-N-SH			child	4 y	cell line	

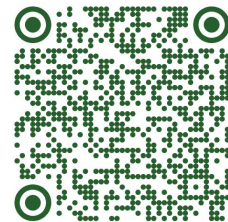
Move

Deselect all - Select all

Search...

- blood
- bone marrow
- brain
- connective tissue
- embryonic
- endocrine gland
- epithelium
- gonad
- heart
- immune organ
- intestine
- kidney
- limb

Regulatory Features tab



regulation | Human > Epigenomes Genome view Regulatory features Source data | Help & Docs About | Release: Latest ⓘ About the ENSEMBL project

GRCh38

Search...

15 entries per page Showing 1 to 15 of 643528 features Download

ID	Type	Location	Genes
ENSR00001408561	EMAR	1:9,953-11,649	
ENSR1_958	Promoter	1:10,936-11,436	
ENSR1_53X2	CTCF	1:11,222-11,243	
ENSR1_53X9	CTCF	1:11,280-11,301	
ENSR1_53XJ	CTCF	1:11,339-11,360	
ENSR1_88N	Enhancer	1:11,437-11,649	
ENSR1_CZ	Enhancer	1:12,686-14,671	
ENSR00001408565	EMAR	1:12,686-14,671	
ENSR1_B48	Enhancer	1:15,101-15,853	

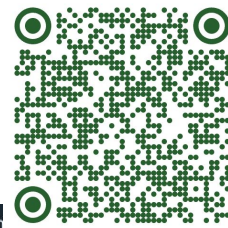
GRCh38

Search promoter

15 entries per page Showing 1 to 15 of 35983 features Download

ID	Type	Location	Genes
ENSR1_958	Promoter	1:10,936-11,436	ENSG00000290825
ENSR1_9B7	Promoter	1:28,220-28,599	ENSG00000243485
ENSR1_8M3	Promoter	1:29,333-29,564	ENSG00000310526, ENSG00000243485
ENSR1_9BN	Promoter	1:29,777-30,277	ENSG00000243485
ENSR1_73K8	Promoter	1:36,436-36,536	ENSG00000308361
ENSR1_843W	Promoter	1:96,548-96,720	ENSG00000308314
ENSR1_93XG	Promoter	1:181,117-181,517	ENSG00000241860, ENSG00000308415

Regulatory Features tab



Click to visualise location

GRCb38 regulatory features / ENSR1_9B7

Classification 1: Promoter

Location 1: 28,220-28,599

Genes MIR1302-2HG

Activity 3 epigenomes

A feature is considered *Active* in an epigenome if it overlaps an open chromatin peak. It is considered *Inactive* when open chromatin data are available but no overlapping peak is detected.

Active (0/3)

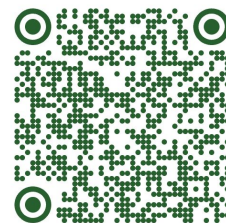
Inactive (3/3)

No open chromatin data (1/3)

astrocyte
brain (120-122 dpf)
neural progenitor cell (f, 5 dpf)

SK-N-SH (f, 4 y)

Genome view tab



regulation | Human > Epigenomes | Genome view | Genes | Regulatory features | Data access | Help & Docs | About | Release: Latest | About the ENSEMBL project

GRCh38

View this location in 1: 28,220-28,599

Search coordinates

Enter a gene, feature or location... chromosome 2 Mbp 100 kbp 1 kbp

Current location

Zoom tools

Genes

GENCODE V48 (Basic)

WASH7P

Regulatory Features

28,300 28,400 28,500

Promoter Enhancer Open chromatin EMAR Epigenetically-modified accessible region CTCF binding site

Genes 2

Regulatory features card

Regulatory features 2

Regulatory activity 4 epigenomes

Filtered epigenomes

Epigenomes

Open chromatin: Peak Open chromatin: Signal H3K4me3 H3K27ac H3K4me1 H3K27me3

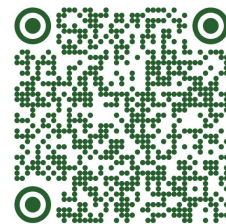
astrocyte

brain (120-122 dpt)

neural progenitor cell (f. 5 dpt)

SK-N-SH (t. 4 y)

Genome view tab: zoomed out



GRCh38

Current coordinates

1: 1-100,000

100 kbp

Enter a gene, feature or location...

chromosome

2 Mbp

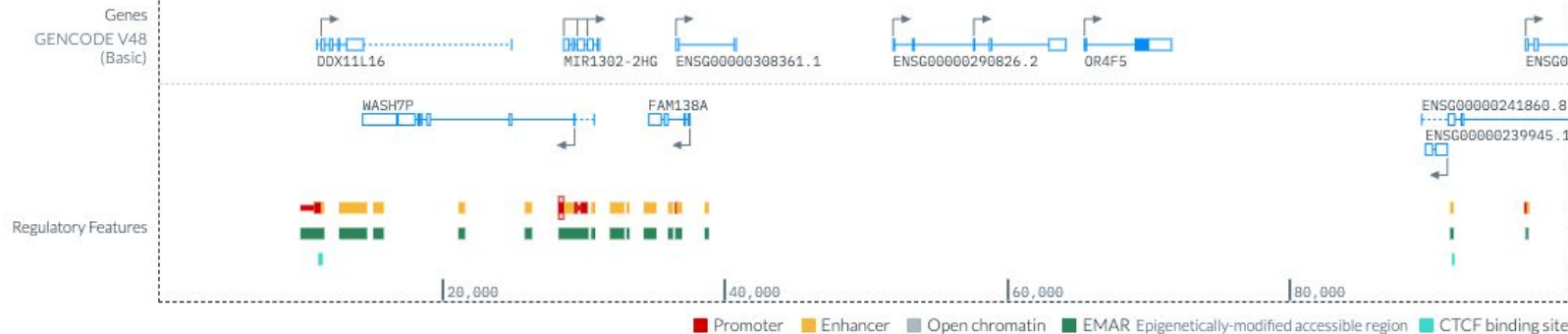
100 kbp

1 kbp

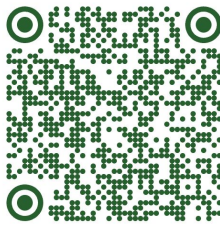


Scale

Zoom out



Genome view tab: chromosome change



GRCh38

Change chromosome

1: 28,220–28,599



Sequence ID

Search...

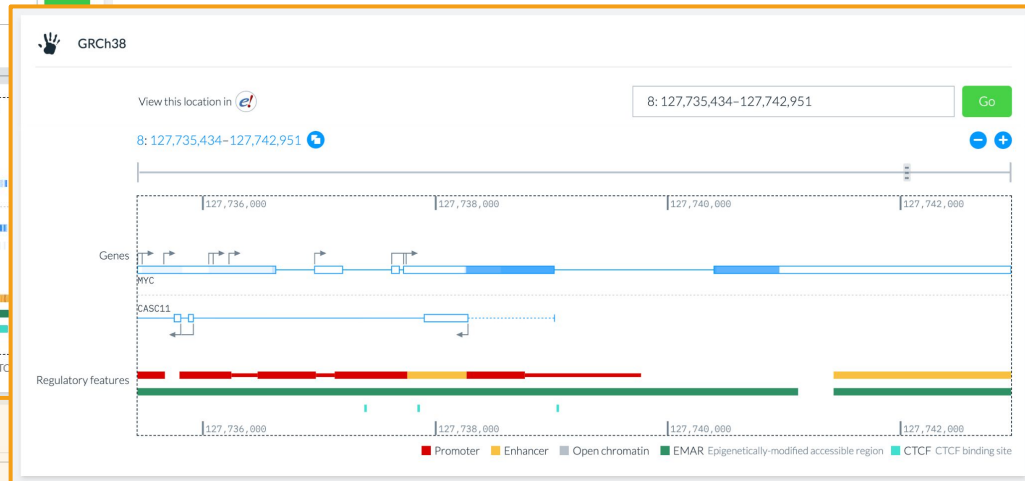
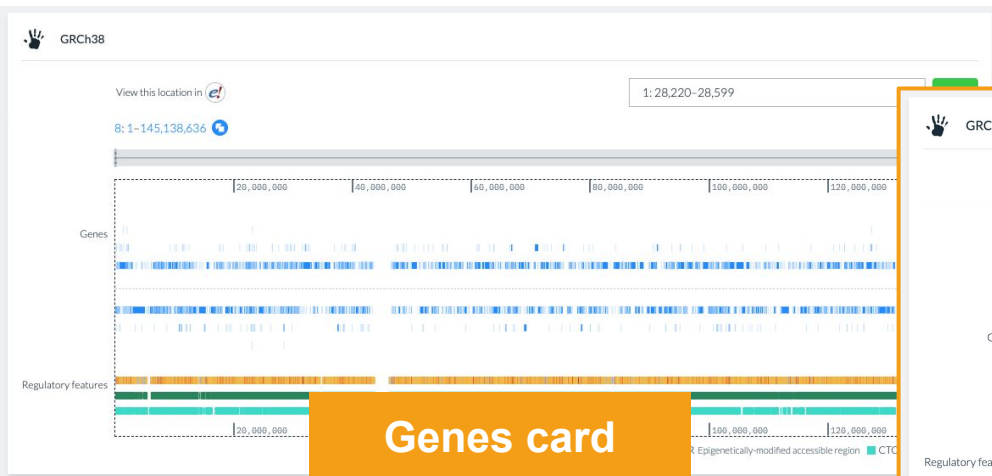
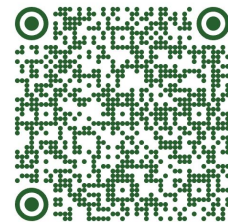
1	14
2	15
3	16
4	17
5	18
6	19
7	20
8	21
9	22

Regul

28,300

28,300

Genome view tab: Gene search



Genes 2305

Search MYC

15 entries per page

Showing 1 to 1 of 1 genes.

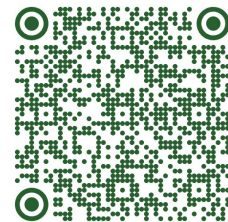
ID	Symbol	Biotype	Location
ENSG00000136997.22	MYC	protein_coding	8: 127,735,434-127,742,951

Regulatory features 30668

Regulatory activity 144 epigenomes

Details can be viewed when the region is less than 2 Mbp.

Genome view tab: Gene page link



GRCh38

View this location in

8: 127,735,434–127,742,951

chromosome 2 Mbp 100 kbp 1 kbp 7.52 kbp

Chromosome 8

Genes
GENCODE V48 (Basic)
MYC
CASC11

Regulatory Features
Promoter Enhancer Open chromatin EMAR Epigenetically-modified accessible region CTCF binding site

Genes 2 **MYC (Protein coding)** X

MYC
Biotype Protein coding
Strand Forward
Location 8: 127,735,434–127,742,951
[Activity summary](#)

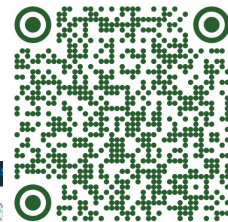
Showing 1 to 2 of 2 genes.

Biotype	Location	Name
lncRNA	8: 127,686,343–127,739,022	cancer susceptibility 11
Protein coding	8: 127,735,434–127,742,951	MYC proto-oncogene, bHLH transcription factor

ENSG00000136997.22

Gene page link

Genes tab



GRCh38 genes / MYC

GENCODE v48 (basic)

ID ENSG00000136997.22

Symbol MYC

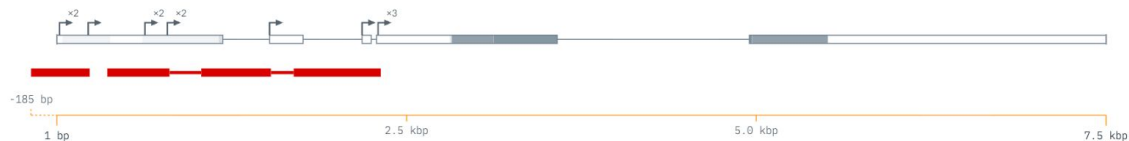
Name MYC proto-oncogene, bHLH transcription factor (HGNC)

Location 8:127,735,434-127,742,951

Biotype Protein coding

Strand Forward

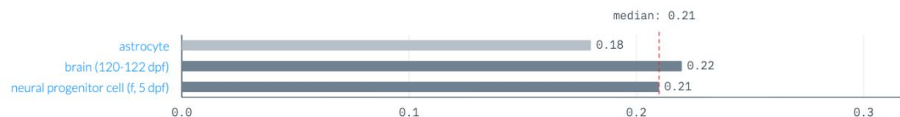
View in



Gene activity 4 epigenomes

This view serves as a guide to gene activity, but it is provisional and details may change. Activity measurements for selected epigenomes are proportional to the overlap between open chromatin signal and the gene's promoters regions, normalised so that the maximum value across the genome is 1.0. Greater promoter accessibility suggests a higher likelihood of gene activity, although it does not directly measure gene expression.

Open chromatin activity (3/4)



No open chromatin data (1/4)

SK-N-SH (f, 4 y)

Genome view tab: Regulatory feature selection

GRCh38

View this location in [ENSEMBL](#)

8:127,731,826-127,746,559 [Link](#)

chromosome 2 Mbp 100 kbp 1 kbp 14.7 kbp

Chromosome 8

Genes
GENCODE V48 (Basic)

MYC

CASC11

Regulatory Features

Regulatory features selection

Genes 2 MYC (Protein-coding)

Regulatory features 15 ENSR8_93TNWH X

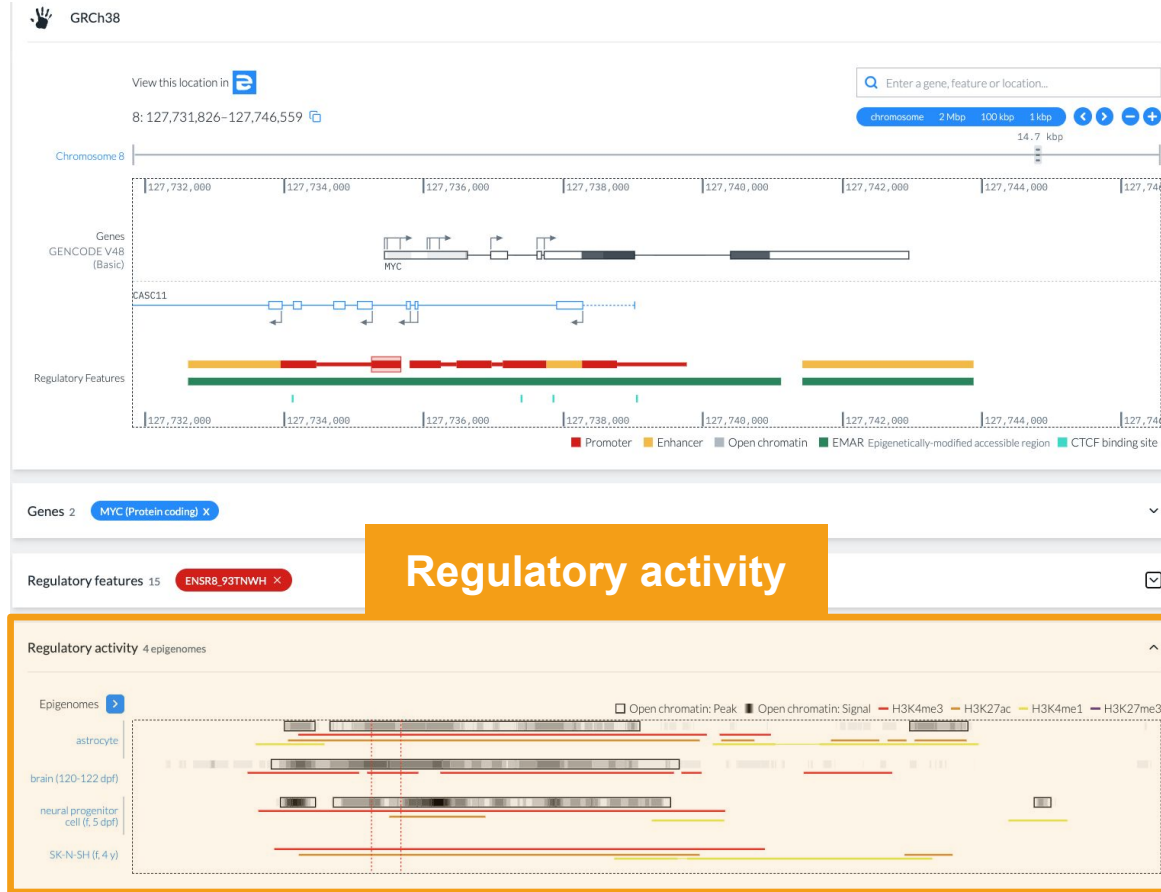
Search...

15 entries per page Showing 1 to 6 of 6 features.

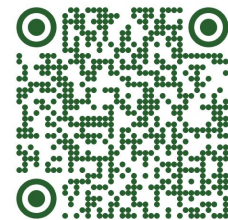
ID	Type	Location	Genes
ENSR8_BXCG3	Promoter	8:127,733,946-127,734,457	CASC11
ENSR8_93TNWH	Promoter	8:127,735,249-127,735,669	CASC11, MYC
ENSR8_93TNWM	Promoter	8:127,735,795-127,736,241	CASC11, MYC
ENSR8_93TNWT	Promoter	8:127,736,468-127,736,968	MYC
ENSR8_BXCGJ	Promoter	8:127,737,131-127,737,755	MYC
ENSR8_93TNXD	Promoter	8:127,738,267-127,738,767	CASC11

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Genome view tab: Regulatory activity



Genes tab



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GRCh38 GENCODE v48 (basic)

Search...

15 entries per page Showing 1 to 15 of 54582 genes

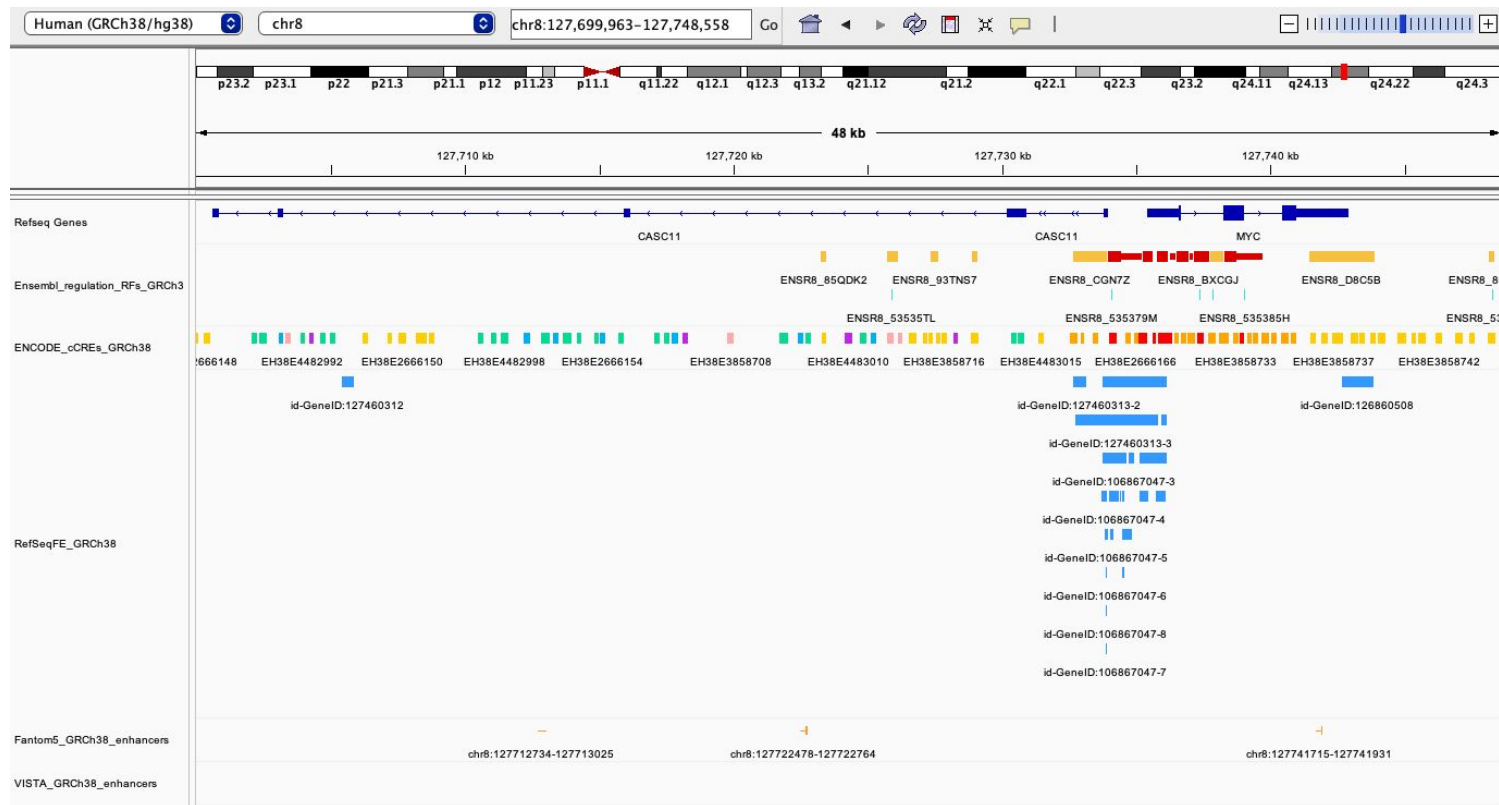
ID	Symbol	Biotype	Location	Name
ENSG00000290825.2	DDX11L16	lncRNA	1:11,121-24,894	DEAD/H-box helicase 11 like 16 (pseudogene)
ENSG00000310526.1	WASH7P	lncRNA	1:14,356-30,744	WASP family homolog 7, pseudogene
ENSG00000243485.6	MIR1302-2HG	lncRNA	1:28,589-31,109	MIR1302-2 host gene
ENSG00000237613.3	FAM138A	lncRNA	1:34,553-37,595	family with sequence similarity 138 member A
ENSG00000308361.1		lncRNA	1:36,526-40,778	
ENSG00000290826.2		lncRNA	1:51,891-64,116	
ENSG00000186092.7	OR4F5	Protein coding	1:65,419-71,585	olfactory receptor family 4 subfamily F member 5
ENSG00000241860.8		lncRNA	1:89,295-181,719	
ENSG00000239945.1		lncRNA	1:89,551-91,105	
ENSG00000308314.1		lncRNA	1:96,638-105,742	
ENSG00000308579.1		lncRNA	1:134,914-136,246	
ENSG00000239906.1		lncRNA	1:139,790-140,339	
ENSG00000310528.1		lncRNA	1:146,386-199,861	
ENSG00000241599.1		lncRNA	1:160,446-161,525	
ENSG00000308415.1	DDX11L2	lncRNA	1:180,931-184,937	DEAD/H-box helicase 11 like 2 (pseudogene)

1

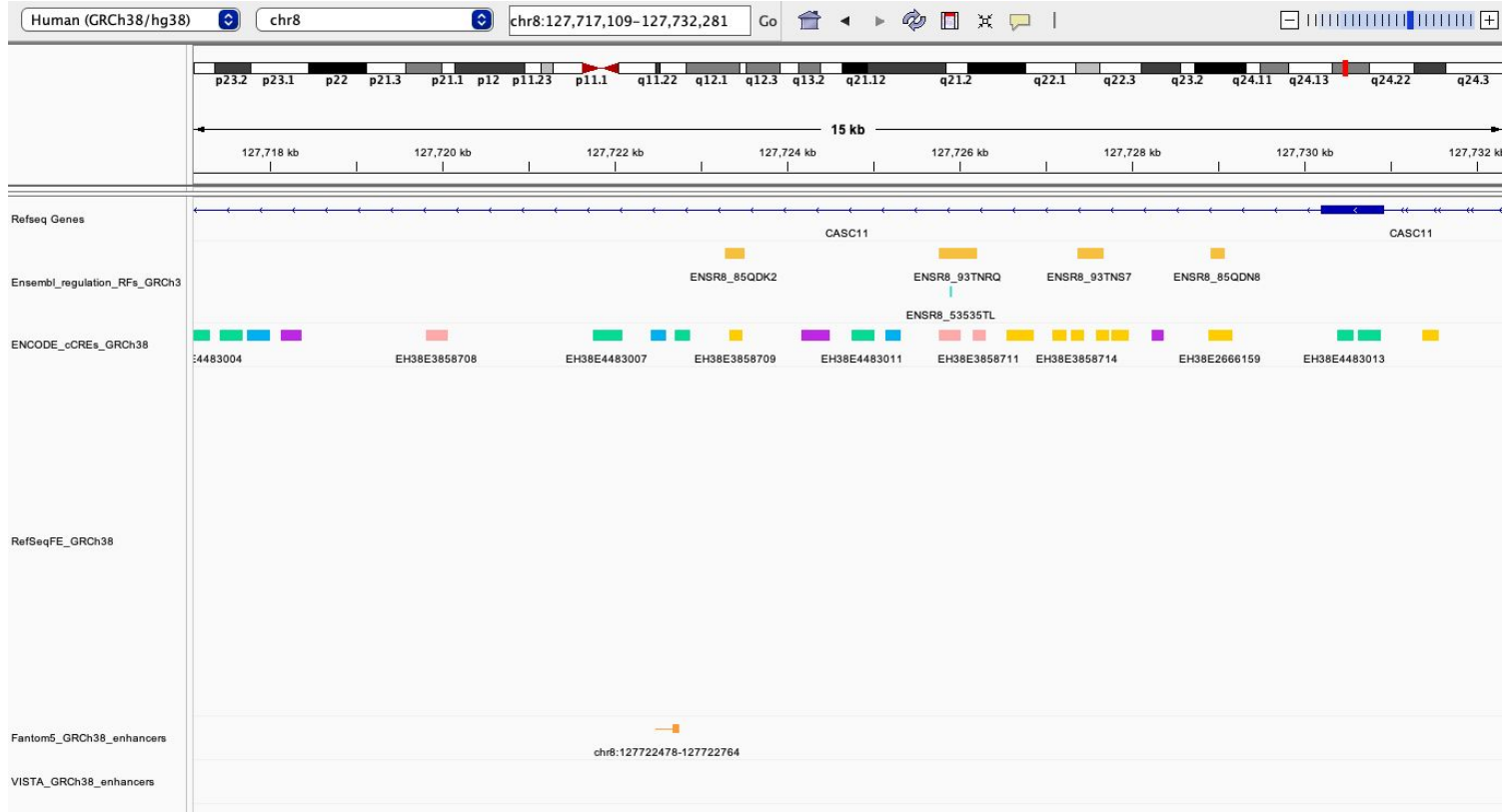
Differences between regulatory annotation frameworks

Resource	Main evidence categories	What it is useful for	Main caveat
<i>Ensembl Regulation</i>	Chromatin accessibility, histone modifications, CTCF ChIP-seq	Integrated regulatory features: promoters, enhancers, open chromatin regions, CTCF sites; activity across epigenomes	Evidence-based annotation, not direct proof of regulatory function
<i>ENCODE cCREs / SCREEN</i>	Chromatin accessibility, histone modifications, CTCF/TF ChIP-seq	Candidate cis-regulatory elements with promoter-like, enhancer-like, CTCF-like and other classes	Strong genome-wide catalogue, but still “candidate” elements
<i>RefSeq Functional Elements</i>	Manually curated experimental evidence, literature, protein binding, functional assays	High-confidence functional regions such as enhancers, silencers, insulators and locus control regions	Curated and evidence-rich, but less comprehensive genome-wide
<i>FANTOM5</i>	Transcription-based evidence, mainly CAGE	Active promoters and transcribed enhancers across many cell types/tissues	Best for active transcribed elements; may miss poised or non-transcribed CREs
<i>Roadmap Epigenomics / ChromHMM</i>	Chromatin accessibility, histone modifications	Chromatin-state annotations across many tissues/cell types	State labels are model-based and assay-dependent. Older technologies
<i>VISTA Browser</i>	Comparative genomics plus in vivo reporter assays	Experimentally validated developmental enhancers	Very valuable but focused on tested fragments, not comprehensive

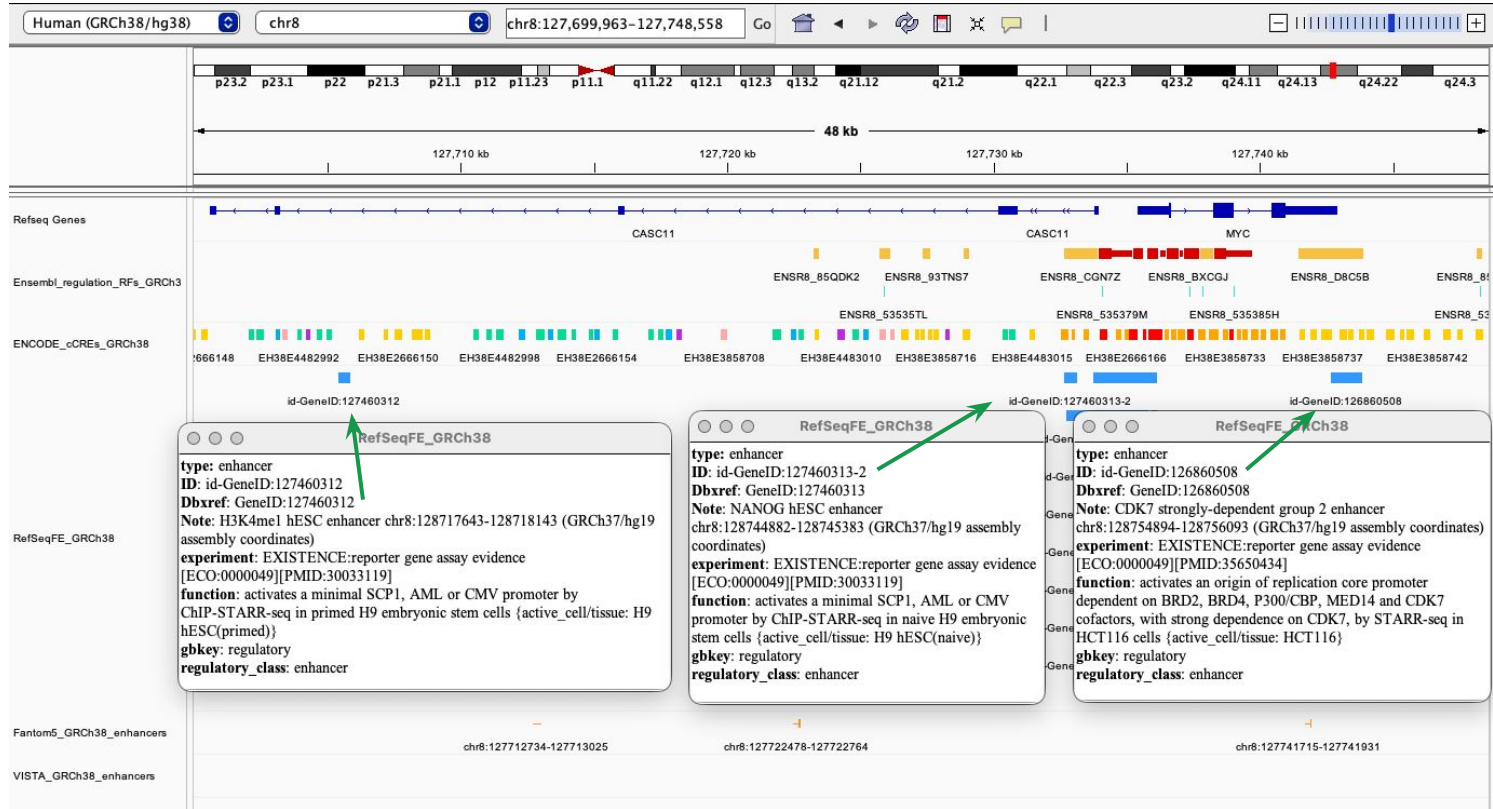
Differences between regulatory annotation frameworks: GRCh38



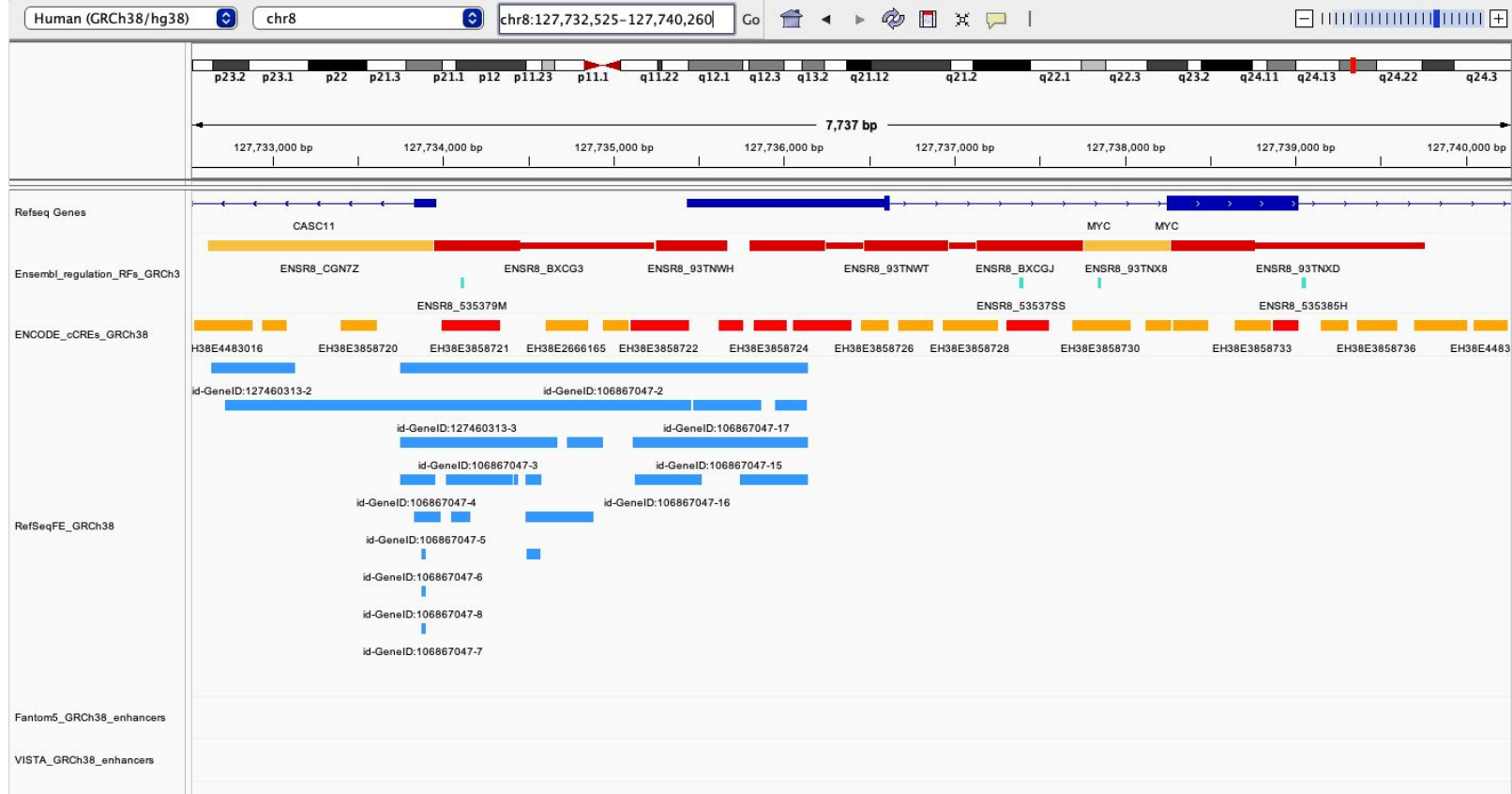
Differences between regulatory annotation frameworks: GRCh38



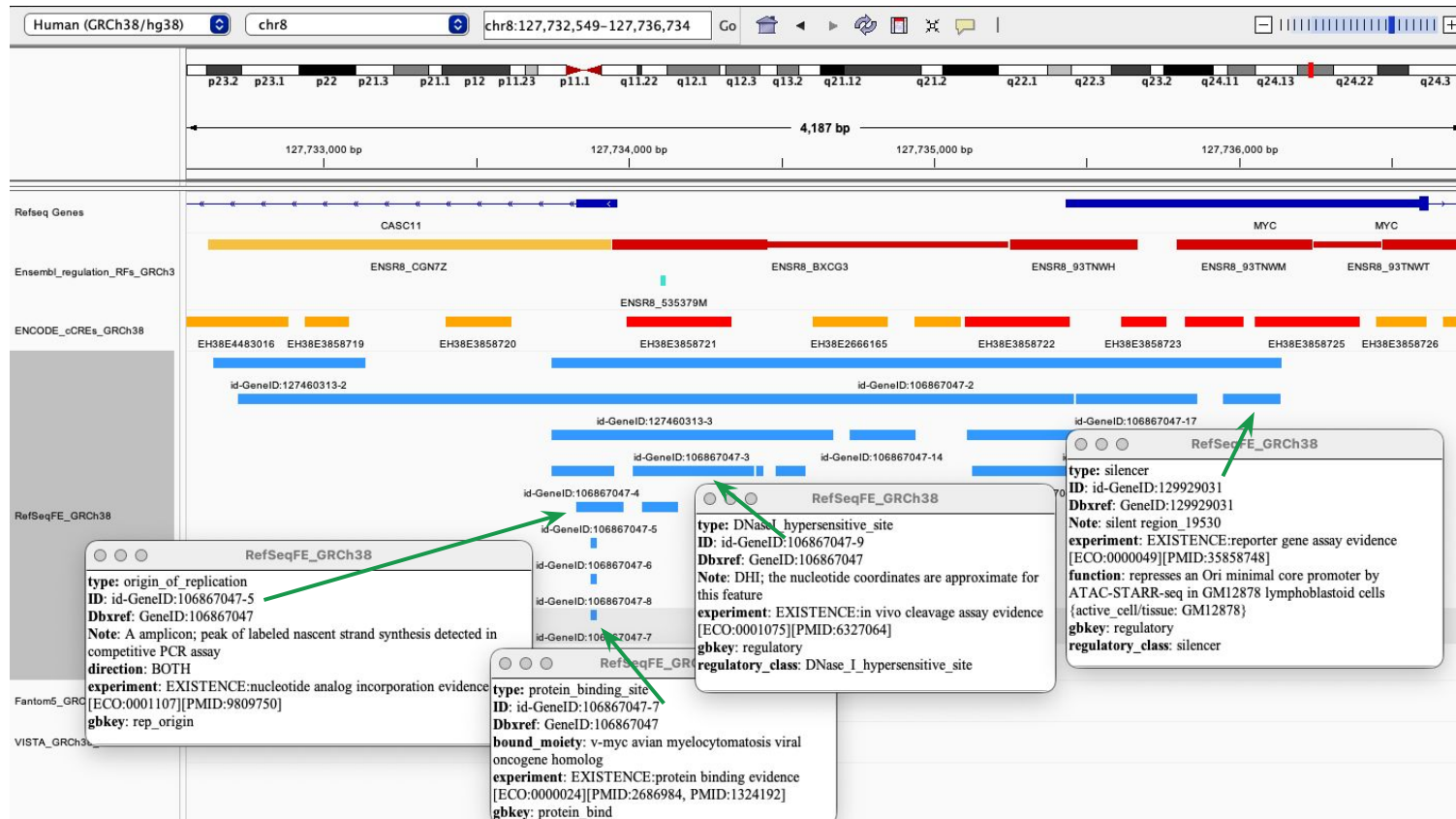
Differences between regulatory annotation frameworks: GRCh38



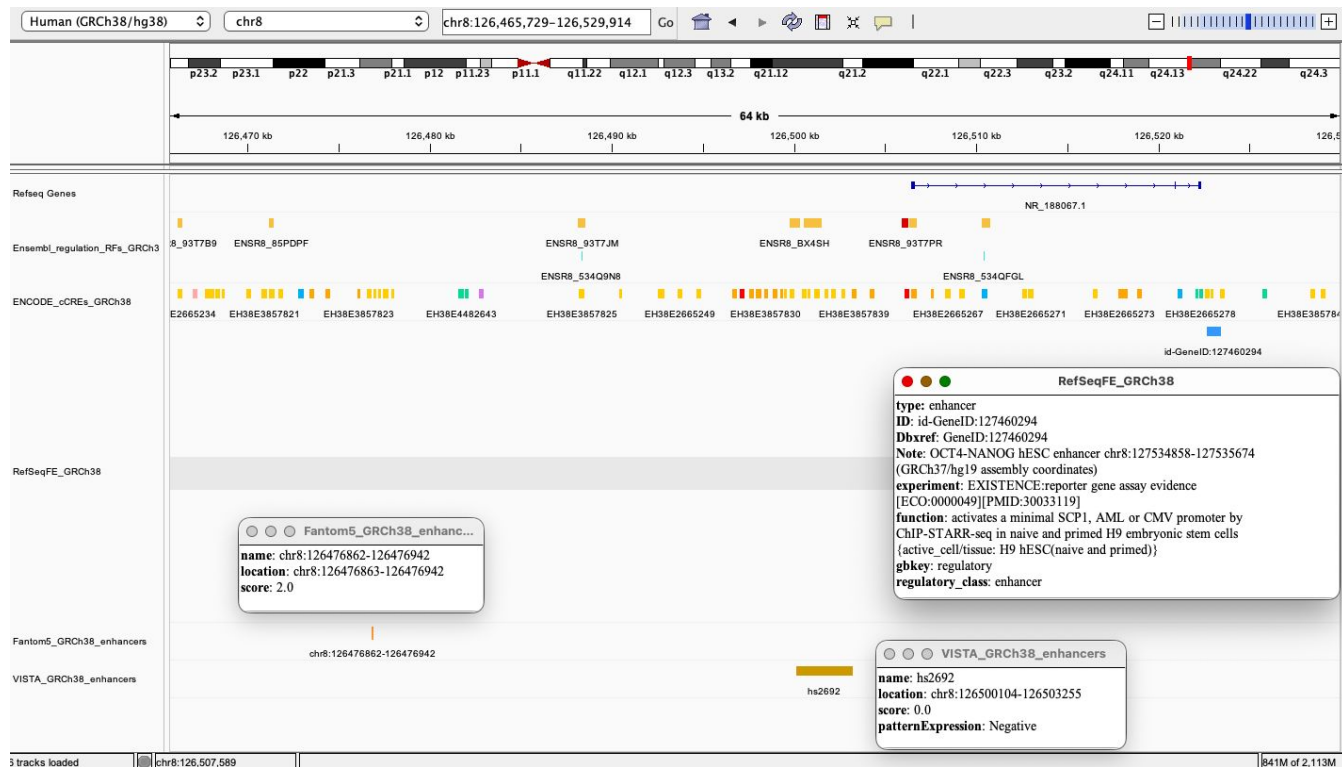
Differences between regulatory annotation frameworks: GRCh38



Differences between regulatory annotation frameworks: GRCh38



Differences between regulatory annotation frameworks: GRCh38



Questions?



Hands-on



Ensembl Regulation

Answer the following questions:

- ✓ Explore the **indicated locus** on the Genome View tab:
 - GRCh38, chr8:127,682,026-127,744,101
 - GRCm39/mm39, chr6:6,740,753-6,891,103
- ✓ Display and explore regulatory features using the Regulation card.
- ✓ Filter epigenomes based on the given criteria
- ✓ Compare regulatory activity across the group of selected epigenomes:
 - GRCh38: Caco-2 and HCT116 cell lines
 - GRCm39: Forebrain epigenomes
- ✓ Inspect supporting epigenomic evidence in the indicated epigenome
- ✓ Identify at least one context-dependent example.
- ✓ Compare the results with the ones obtained in the previous session.